

**INTELLIGENT TRAFFIC MANAGEMENT USING COMPUTER  
VISION  
INNOVATIVE DESIGN LAB – II**

A Report

Submitted by

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in

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**SRI SAIRAM ENGINEERING COLLEGE**

(An Autonomous Institution; Affiliated to Anna University, Chennai-600044)

**ANNA UNIVERSITY: CHENNAI 600 044**

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## **SRI SAIRAM ENGINEERING COLLEGE**

(An Autonomous Institution; Affiliated to Anna University, Chennai -600 044)

### **BONAFIDE CERTIFICATE**

Certified that this Innovative Design Lab-II report "INTELLIGENT TRAFFIC MANAGEMENT USING COMPUTER VISION" is the bonafide work of "KAVISH S 2403412520521034, HARITHARAN S 2403412520521025, KAVINILAVAN S 2403412520521033" who carried out the (24ITID401 – INNOVATION DESIGN LABORATORY - II) under my supervision.

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**INTERNAL EXAMINER**

**EXAMINER**

**EXTERNAL**

## **ACKNOWLEDGEMENT**

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# **ABSTRACT**

## **INTELLIGENT TRAFFIC MANAGEMENT USING COMPUTER VISION**

This report presents the systematic development and feasibility evaluation of an AI-driven intelligent traffic management platform designed for Indian urban municipalities. The project follows a structured entrepreneurial framework encompassing problem identification, customer validation, solution ideation, opportunity mapping, prototype development, business modelling, financial planning, and feasibility assessment. The proposed solution leverages real-time data from IoT sensors, traffic cameras, and GPS systems to dynamically optimize traffic signal timings, predict congestion patterns, and create prioritized green corridors for emergency vehicles. The system is delivered as a Software-as-a-Service (SaaS) platform built on Python with TensorFlow for AI modelling, React and Node.js for dashboard development, and PostgreSQL for data management. The study confirms the technical viability, market demand, scalability potential, and financial sustainability of the venture, aligning with SDG 11: Sustainable Cities and Communities.

# CHAPTER 1

## ORIENTATION & ENTREPRENEURIAL FOUNDATION

### 1.1 Entrepreneurial Leadership Insights

Through the Wadhvani Foundation orientation modules, our team developed a foundational understanding of entrepreneurship as a process of identifying problems and creating measurable value for society. Key insights included the distinction between a job-seeking mindset and an entrepreneurial mindset — where entrepreneurs embrace uncertainty, take calculated risks, and iterate based on feedback rather than waiting for perfect conditions.

The orientation helped us cultivate leadership qualities including resilience (continuing to refine our traffic management concept despite initial scepticism from stakeholders), adaptability (pivoting from a hardware-first approach to a SaaS model based on market feedback), and accountability (setting clear milestones and ownership within our three-member team). We learned that entrepreneurial success is less about a single breakthrough idea and more about disciplined execution, customer empathy, and consistent value delivery.

### 1.2 Design Thinking Approach

The Design Thinking framework was central to how we approached the Intelligent Traffic Management problem:

- Empathize: We conducted five stakeholder interviews with business professionals, daily commuters, and a traffic department official. We observed real-time traffic patterns at key Chennai intersections and gathered survey responses from 30+ participants to understand pain points beyond surface-level frustration.
- Define: From the empathy stage, we crystallized the core problem — urban traffic management systems in India rely on fixed-time signals that cannot adapt to real-time conditions, causing congestion, economic losses, and delayed emergency response.
- Ideate: We brainstormed over 15 solution concepts using mind mapping and SCAMPER techniques, ranging from physical infrastructure upgrades to mobile-

based routing, ultimately selecting an AI-driven adaptive signal control platform for its scalability and non-invasiveness.

- **Prototype:** We developed a digital interactive MVP simulating the core dashboard interface with live traffic flow visualization, adaptive signal control simulation, and one-click green corridor activation.
- **Test:** The MVP was tested with 10 potential users including traffic officials and urban planners, yielding actionable feedback on UI intuitiveness and AI accuracy requirements.

### **1.3 Vision & Mission**

- **Vision:** To become the leading AI-powered traffic management solution for smart cities globally, enabling safer, more efficient urban mobility within the next decade.
- **Mission:** To eliminate traffic congestion, enhance public safety through faster emergency response, and promote sustainable urban development through data-driven, real-time traffic optimization for Indian metropolitan cities.

### **1.4 Overview of Business Idea**

The project addresses the critical challenge of urban traffic congestion in India's high-density cities. The proposed solution is a cloud-based AI platform — referred to as Nameless — that replaces static traffic signal systems with dynamic, real-time adaptive control. By integrating live data from city cameras, IoT sensors, and GPS feeds, the system intelligently manages traffic signals, predicts congestion hotspots, and creates automatic green corridors for emergency vehicles. The platform targets municipal traffic authorities as its primary customers, delivered via a SaaS subscription model that eliminates large upfront capital expenditure for cities.

## **CHAPTER 2**

### **PROBLEM IDENTIFICATION**

#### **2.1 Macro Industry–Problem Perspective**

India's urban transportation sector is under severe stress. With over 500 million people living in cities and rapid urbanization continuing at approximately 2.3% annually, the demand on urban road infrastructure has far outpaced its capacity. The intelligent transportation systems (ITS) industry globally is valued at over USD 28 billion and is growing at a CAGR of 8–10%, driven by the convergence of IoT, AI, and 5G connectivity. Yet, Indian cities continue to rely on legacy fixed-time traffic signal systems designed for traffic volumes of the 1990s.

Government policy is beginning to catch up — the Smart Cities Mission (launched 2015) has allocated over Rs. 2 lakh crore across 100 cities for digital infrastructure, including intelligent traffic management. However, adoption of modern adaptive traffic systems remains limited due to high procurement costs, lack of local vendors, and lengthy government procurement cycles. This creates a significant technology and service gap that a cost-effective, India-specific SaaS platform can address.

#### **2.2 Real-World Problem Identification**

Field observations at key Chennai intersections (Anna Salai, Kathipara, Koyambedu) revealed that traffic signals operate on pre-programmed fixed cycles even during off-peak hours, resulting in unnecessary vehicle idling and fuel waste. Interviews with a traffic department official confirmed that manual overrides are the only current mechanism for handling non-standard situations. Analysis of ambulance dispatch records (Times of India, 2023) showed that emergency vehicle response times in Chennai had increased by up to 75% due to traffic delays — directly impacting patient survival outcomes. A survey of 30 respondents showed that 87% experience daily commute delays of over 30 minutes, and 93% believe current traffic management is inadequate.

#### **2.3 Problem Statement**

"The problem of fixed-time, non-adaptive traffic signals affects urban commuters, emergency responders, and municipal authorities in Indian metropolitan cities due to the

absence of real-time traffic data integration and intelligent control mechanisms, resulting in severe congestion, increased commute times, higher fuel consumption, and critically delayed emergency vehicle response times."

## **2.4 Problem Analysis**

Root Cause Analysis (5 Whys):

- Why is there traffic congestion? → Signals do not adapt to actual traffic volumes.
- Why don't signals adapt? → They operate on fixed pre-programmed timers.
- Why are fixed timers used? → No real-time data collection or processing infrastructure exists.
- Why is there no real-time infrastructure? → High cost of hardware and lack of local AI-based solutions.
- Why are local solutions absent? → Limited intersection between urban planning priorities and advanced technology adoption in Indian municipal procurement.

Contributing factors include: outdated procurement frameworks, lack of skilled traffic management personnel, absence of centralized monitoring dashboards, and insufficient inter-agency data sharing between traffic police, municipal corporations, and emergency services.

## **2.5 Problem Interviews & Survey Results**

Five in-depth stakeholder interviews were conducted with business professionals, self-employed individuals, and a government traffic official. All five expressed dissatisfaction with existing traffic management. Four out of five rated the problem as critically important. A broader survey of 30 respondents confirmed: 87% face daily commute delays exceeding 30 minutes, 93% find current management inadequate, and 80% expressed strong interest in an AI-powered solution. These findings validate the existence of a significant, widespread, and underserved problem warranting a technology-driven solution.



## **CHAPTER 3**

### **JUSTIFICATION FOR SDG ALIGNMENT**

#### **3.1 Relevant SDG Goals Identified**

The project primarily aligns with SDG 11: Sustainable Cities and Communities, and secondarily with SDG 3: Good Health and Well-being, and SDG 13: Climate Action.

#### **3.2 Rationale for SDG Selection**

SDG 11 is directly linked because the platform improves urban mobility infrastructure, which is a core component of making cities inclusive, safe, resilient, and sustainable. SDG 3 is relevant because faster emergency vehicle response times directly save lives. SDG 13 is relevant because reduced vehicle idling at intersections decreases fuel consumption and vehicular emissions — a meaningful contribution to urban carbon footprint reduction.

#### **3.3 Mapping Problem to SDG Targets**

SDG 11, Target 11.2: Provide access to safe, affordable, accessible, and sustainable transport systems for all, improving road safety — directly addressed through adaptive signal control and emergency corridors. SDG 3, Target 3.6: Halve the number of global deaths and injuries from road traffic accidents — supported by faster emergency response capability. SDG 13, Target 13.2: Integrate climate change measures into policies and planning — addressed through fuel consumption reduction from optimized signal timing.

#### **3.4 Social, Economic & Environmental Impact**

- **Social Impact:** Reduced commute times improve quality of life; faster emergency response saves lives; improved traffic flow benefits pedestrians and cyclists.
- **Economic Impact:** Cities save on fuel costs; reduced congestion increases economic productivity; SaaS model creates employment in AI/cloud engineering; payback period of 6 months for municipal customers.
- **Environmental Impact:** Optimized signal timing reduces vehicle idling, lowering CO2 emissions; supports India's commitment to reduce carbon intensity under the Paris Agreement.

### **3.5 Long-Term Sustainability Contribution**

The platform creates sustainable value by continuously learning from traffic data, improving its predictions over time without requiring hardware replacements. Its cloud-native, modular architecture ensures it can scale to additional cities and integrate with future smart city infrastructure (autonomous vehicles, V2X communication). The SaaS model ensures long-term financial viability for both the vendor and municipal customers, avoiding the pitfalls of one-time capital expenditure projects that lack ongoing support.

## **CHAPTER 4**

### **CUSTOMER IDENTIFICATION & NEED VALIDATION**

#### **4.1 Customer Segmentation**

Primary Customers: City traffic engineering departments and urban municipal transport authorities in Indian metropolitan cities with populations exceeding 5 million (Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Pune). These are B2G (business-to-government) customers with defined procurement processes.

Secondary Customers: Private infrastructure developers managing smart townships, special economic zones, and airport access roads. These are B2B customers with faster procurement cycles and willingness to pay premium pricing.

#### **4.2 Jobs-To-Be-Done (JTBD)**

- **Functional Jobs:** Manage traffic flow at intersections in real time, reduce peak-hour congestion, ensure ambulances and fire services reach destinations faster, generate traffic data reports for urban planning.
- **Emotional Jobs:** Reduce public criticism directed at traffic departments, demonstrate technological innovation to political leadership, feel confident that the city's infrastructure is modern and data-driven.
- **Social Jobs:** Be seen as a forward-thinking, citizen-responsive government department; meet Smart City Mission benchmarks; attract positive media coverage.

#### **4.3 Customer Persona**

Name: Rajesh Kumar, Age: 42, Occupation: Strategic Planning Manager, Metropolitan Traffic Department, Income: Government Grade A officer. Pain Points: Managing congestion with limited budget and personnel, facing public outrage on social media over traffic delays, dealing with political pressure to show results quickly, concerned about ambulance delays and associated public health implications. Buying Behavior: Responds to government tenders, attends smart city conferences, influenced by successful pilot deployments in peer cities, requires strong ROI justification and vendor credibility.

#### **4.4 Needs Validation Summary**

Survey and interview data confirmed strong demand: all five interviewed stakeholders cited traffic management as a critical problem. 4/5 confirmed they would consider adopting an AI-based solution if demonstrated through a pilot. The broader survey showed 80% interest in an intelligent traffic system. The problem is significant, measurable, and underserved by current solutions, confirming that strong product-market fit potential exists for the proposed platform.

## **CHAPTER 5**

### **SOLUTION IDEA GENERATION**

#### **5.1 Ideation Techniques Used**

The team employed Brainstorming (generating 15+ raw ideas without filtering), Mind Mapping (linking sub-problems like congestion, emergency response, and data collection to solution branches), SCAMPER (Substitute hardware sensors with software-based video analytics; Combine GPS data with camera feeds; Adapt existing signal hardware to accept software commands), and Design Thinking Prototyping (building low-fidelity wireframes before committing to technology choices).

#### **5.2 Alternative Solution Concepts**

Option 1 — Hardware-Based Sensor Grid: Install physical vehicle counters at all intersections. Pros: High accuracy. Cons: Extremely expensive, requires road disruption, long installation timelines, high maintenance.

Option 2 — Mobile Crowdsourcing (Waze-style): Aggregate GPS data from commuter smartphones to reroute traffic. Pros: Low cost, fast deployment. Cons: Does not control signals, passive not active system, limited coverage in low-smartphone-density areas.

Option 3 — AI-Powered Computer Vision Platform (Selected): Use existing city traffic cameras with AI models to count vehicles and control signals in real time. Pros: Leverages existing infrastructure, scalable, no road disruption, active control capability, emergency vehicle detection. Cons: Requires robust AI model, dependent on camera quality and connectivity.

#### **5.3 Final Proposed Solution**

The platform integrates with existing city CCTV cameras and GPS emergency vehicle trackers via cloud APIs. Computer vision models (built with Python and TensorFlow, trained on Indian traffic datasets) analyse live video feeds to count vehicles, estimate congestion density, and detect emergency vehicles. The AI engine dynamically adjusts signal timing at each intersection based on real-time vehicle counts, and automatically creates green corridors (sequential green signals along a route) when an emergency vehicle is detected. A centralized React/Node.js dashboard provides traffic operators with a real-

time city-wide view, predictive congestion alerts, and performance analytics. All data is stored and processed on a PostgreSQL database on a scalable cloud infrastructure.

#### **5.4 Unique Selling Proposition (USP)**

The platform is differentiated by: (1) Real-time AI signal adaptation using existing camera infrastructure — no hardware installation required; (2) Automatic emergency vehicle green corridor creation — a life-saving feature absent in competitor products; (3) India-specific AI models trained on chaotic Indian traffic patterns for superior accuracy; (4) Flexible SaaS pricing that dramatically reduces upfront cost compared to global competitors like Siemens Mobility and SWARCO.

## CHAPTER 6

### OPPORTUNITY & COMPETITION MAPPING

#### 6.1 Industry Trends

The global intelligent traffic management market is growing at approximately 18% CAGR in India between 2024 and 2027, driven by: proliferation of IoT sensors and 5G connectivity enabling real-time data processing; the Indian government's Smart Cities Mission pushing digital transformation in urban infrastructure; rising public demand for efficient and sustainable urban mobility; and increasing availability of low-cost cloud computing making AI-based solutions economically viable for government procurement.

#### 6.2 Global & Local Competitors

Direct Competitors: Siemens Mobility (Sitraffic), SWARCO (UTOPIA), Cubic Transportation Systems (NextTraffic) — all offer traffic management systems but are cost-prohibitive and lack real-time AI adaptability for Indian conditions. Indirect Competitors: Google Maps, Waze — provide traffic data and routing guidance but do not control traffic signals and are consumer-facing rather than infrastructure-facing.

#### 6.3 Competition Analysis

| Feature                          | Our Platform    | Siemens Sitraffic | SWARCO UTOPIA | Google Maps/Waze  |
|----------------------------------|-----------------|-------------------|---------------|-------------------|
| Real-time AI Signal Adaptation   | Yes             | Partial           | Partial       | No                |
| Emergency Vehicle Green Corridor | Yes (Automatic) | Manual            | Manual        | No                |
| India-Specific AI Training       | Yes             | No                | No            | N/A               |
| SaaS / No Hardware Required      | Yes             | No                | No            | N/A               |
| Pricing (per city pilot)         | Rs. 1 Cr        | Rs. 10–50 Cr      | Rs. 8–40 Cr   | Free (no control) |
| Centralized Dashboard            | Yes             | Yes               | Yes           | No                |

SWOT Analysis: Strengths — India-specific AI, low-cost SaaS, emergency corridor feature. Weaknesses — Limited brand recognition, small team. Opportunities — Smart Cities Mission funding, 100+ target cities. Threats — Long government procurement cycles, competition from global incumbents.

## **6.4 Market Size Estimation (TAM, SAM, SOM)**

- Total Addressable Market (TAM): Global intelligent traffic management market — Rs. 23,000 crore by 2028.
- Serviceable Available Market (SAM): Indian metropolitan cities with population >5 million — Rs. 4,600 crore by 2028.
- Serviceable Obtainable Market (SOM): Initial deployment in 2 cities within 2 years — Rs. 16 crore.

## **6.5 Introduction to Financial Planning**

Initial cost estimation includes AI model development (Rs. 50 lakh), cloud infrastructure setup (Rs. 20 lakh), and sales/marketing (Rs. 30 lakh) for Year 1. Revenue is based on a pilot contract price of Rs. 1 crore per city, with recurring annual maintenance of Rs. 20 lakh per city. Pricing is benchmarked at 90% below global competitors, enabling market penetration while maintaining healthy gross margins.



## CHAPTER 7

### PROTOTYPE DEVELOPMENT & MVP

#### 7.1 Prototype Overview

The initial prototype is a digital interactive simulation of the traffic management platform, accessible via web browser. It demonstrates the core system concept without requiring actual camera integration or hardware deployment. The prototype was built using React for the frontend and Node.js for the backend API simulation, with mock data representing live traffic feeds from a four-intersection grid in Chennai. The project repository is available at: <https://haritharan-s.github.io/Nameless-Project/> and a demonstration video at: <https://youtu.be/fPSPeEAfJNs>

#### 7.2 Minimum Viable Product (MVP)

The MVP is a functional software prototype that includes:

- Live traffic flow visualization on a map-based dashboard showing vehicle density per intersection.
- Adaptive signal control simulation — the system adjusts green light duration based on simulated vehicle counts.
- One-click emergency green corridor creation — simulates sequential signal prioritization along a selected route.
- Basic analytics panel showing congestion trends over a simulated 24-hour period.

The MVP was tested in a controlled environment with mock data and demonstrated to 10 potential users including traffic officials and urban planners.

#### 7.3 Product Features & Benefits

| Feature                           | Benefit to Customer                                  |
|-----------------------------------|------------------------------------------------------|
| Real-time adaptive signal control | Reduces average commute times by 15–20%              |
| Emergency vehicle green corridor  | Reduces emergency response time by 25–30%            |
| Predictive congestion alerts      | Allows proactive traffic management                  |
| Centralized dashboard             | Single pane of glass for city-wide traffic oversight |

|                            |                                                 |
|----------------------------|-------------------------------------------------|
| Cloud-native SaaS delivery | No hardware investment; instant updates         |
| India-specific AI models   | Superior accuracy for Indian traffic conditions |

## 7.4 MVP Validation & Feedback

The MVP was demonstrated to 10 users. Seven respondents praised the intuitive visualization and emergency corridor feature, describing it as immediately understandable and practically valuable. Three expressed concerns about real-world AI accuracy for multi-intersection scenarios and integration with legacy traffic hardware. Based on this feedback, the development roadmap was updated to prioritize: (1) real-world camera data integration testing in one live intersection, (2) robust multi-intersection AI model training, and (3) development of a hardware abstraction layer to interface with existing signal controllers.

## CHAPTER 8

### BUSINESS MODELLING

#### 8.1 Lean Canvas

| Block                    | Detail                                                                                                                         |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Problem                  | Fixed-time traffic signals cause congestion, economic losses, and delayed emergency response                                   |
| Customer Segments        | Municipal traffic authorities in Indian metros with population >5 million; private smart city developers                       |
| Unique Value Proposition | AI-powered adaptive signals + automatic emergency corridors, using existing cameras, at 90% lower cost than global competitors |
| Solution                 | Cloud-based AI platform with computer vision, real-time signal control, and SaaS delivery                                      |
| Channels                 | Direct enterprise sales, smart city conferences, government tenders, digital marketing                                         |
| Revenue Streams          | SaaS subscription (annual), pilot project contracts, maintenance contracts                                                     |
| Cost Structure           | AI development, cloud infrastructure, sales & marketing, customer onboarding                                                   |
| Key Metrics              | Cities deployed, commute time reduction %, emergency response improvement %, CSAT score                                        |
| Unfair Advantage         | India-specific AI training data, emergency corridor automation, local market expertise                                         |

#### 8.2 Business Model Overview

The platform operates on a B2G (Business-to-Government) SaaS model. Municipal authorities subscribe to the platform annually, with pricing scaled by the number of intersections managed. The model eliminates large upfront capital expenditure for cities and provides predictable recurring revenue for the company. Optional project-based licensing is available for private infrastructure developers.

#### 8.3 Revenue Model

- Pilot Project Contract: Rs. 1,00,00,000 per city — covers AI model customization, system integration with existing cameras and signal hardware, staff training, and 12-month support.
- Annual Maintenance & Support: Rs. 20,00,000 per city per year — includes model retraining, platform updates, and 24x7 support.
- Premium Feature Licensing: Rs. 10,00,000 per year — for advanced analytics, predictive modelling dashboards, and API access for third-party integration.

## **8.4 Sales & Go-To-Market Strategy**

Phase 1 (Months 1–3): Secure one municipal pilot through direct engagement with Chennai or Bengaluru traffic engineering department, leveraging Smart Cities Mission procurement channels. Phase 2 (Months 4–6): Use pilot success metrics as case study for second city contract. Phase 3 (Months 7–12): Expand to 4 cities, establish government framework agreement for standardized procurement.

Sales channels include direct enterprise engagement, partnerships with smart city system integrators (e.g., L&T Smart World & Communication, Tata Projects), and participation in government tender portals (GeM, CPP).

## **8.5 Marketing & Customer Acquisition Strategy**

The marketing strategy emphasizes the platform's life-saving emergency corridor capability as the lead value proposition. Targeted campaigns on LinkedIn directed at traffic department heads and smart city programme managers will build awareness. Case studies from pilot deployments, peer city referrals, and invitations to demonstrate at Smart Cities India Expo will drive credibility. Customer acquisition is further supported by offering free 3-month pilot evaluations to reduce procurement risk for municipalities.

## CHAPTER 9

### FINANCIAL MANAGEMENT & PROFITABILITY

#### 9.1 Financial Overview

Financial projections are presented in Indian Rupees (INR) and based on a SaaS service-provider model targeting municipal traffic authorities. Startup costs are primarily allocated to AI development (50%), cloud infrastructure (20%), and sales & marketing (30%). The company requires initial seed funding of Rs. 2 crore to sustain the first 12 months of operations.

#### 9.2 Path to Profitability

The business is projected to reach break-even by Month 3 of Year 2, following pilot revenue realization in Year 1. Profitability is achieved by end of Year 2 through expansion to multiple city clients and upselling of premium analytics features. The relatively low marginal cost of adding new city clients (primarily AI model customization and onboarding) enables rapid gross margin improvement as the customer base scales.

#### 9.3 Forecasted Profit & Loss (P&L)

| Year   | Revenue (INR) | Total Expenses (INR) | Profit / (Loss) (INR) |
|--------|---------------|----------------------|-----------------------|
| Year 1 | 1,80,00,000   | 2,20,00,000          | (40,00,000)           |
| Year 2 | 4,50,00,000   | 3,50,00,000          | 1,00,00,000           |
| Year 3 | 10,00,00,000  | 6,50,00,000          | 3,50,00,000           |

#### 9.4 Financial Projections

Revenue growth is driven by expansion to 2 cities in Year 1 (pilot contracts), 4 cities in Year 2 (full deployments + recurring maintenance), and 8 cities in Year 3 (scaled rollout + premium feature upsells). Cloud infrastructure costs scale sub-linearly with new city deployments, improving unit economics over time.

#### 9.5 Unit Economics

- Customer Acquisition Cost (CAC): Rs. 1,976
- Customer Lifetime Value (CLV): Rs. 10,776
- CLV/CAC Ratio: 5.5x (highly favourable — industry benchmark is >3x)
- Payback Period: 6 months
- Gross Margin (Year 2 onwards): ~55%

## **9.6 Funding Plan & Capital Requirements**

An initial seed funding of Rs. 2 crore is sought to support the first 12 months. Allocation:

- 50% (Rs. 1 crore): Technology development — AI model training, platform engineering, cloud infrastructure setup
- 30% (Rs. 60 lakh): Sales and marketing — conference participation, demo video production, enterprise sales team
- 20% (Rs. 40 lakh): Operational expenses — team salaries, legal, administrative

Funding sources being explored include Startup India Seed Fund Scheme (SISFS), NASSCOM's 10,000 Startups programme, and angel investment from urban mobility sector investors.

## CHAPTER 10

### TEAM & EXECUTION

#### 10.1 Team Composition and Roles

- **KAVISH S (Team Lead):** Oversees project management, business development, investor relations, and stakeholder communication. Responsible for go-to-market strategy execution and government liaison.
- **HARITHARAN S (Technical Lead):** Responsible for AI/ML model development, computer vision pipeline, system integration with traffic cameras and signal controllers, and cloud backend architecture.
- **KAVINILAVAN S (Communication & Outreach Lead):** Handles client interactions, platform documentation, partnership development, marketing content creation, and pilot deployment coordination.

#### 10.2 Milestones & Execution Plan

| Timeline     | Milestone                                                                                                        |
|--------------|------------------------------------------------------------------------------------------------------------------|
| Months 1–3   | Secure pilot agreement with one city; refine AI model with real camera data; complete system integration testing |
| Months 4–6   | Go-live on pilot city; collect performance data; secure second city contract                                     |
| Months 7–9   | Publish pilot case study; present at Smart Cities India Expo; begin second city deployment                       |
| Months 10–12 | Reach profitability run-rate; deploy in four cities; establish government framework agreement                    |
| Year 2       | Scale to 8 cities; launch premium analytics module; explore Southeast Asia expansion                             |

#### 10.3 Risk Analysis and Mitigation Strategies

| Risk                                  | Likelihood | Impact | Mitigation Strategy                                                        |
|---------------------------------------|------------|--------|----------------------------------------------------------------------------|
| Lengthy government procurement cycles | High       | High   | Engage system integrators; offer free pilots to demonstrate value quickly; |

|                                                   |        |        |                                                                                                      |
|---------------------------------------------------|--------|--------|------------------------------------------------------------------------------------------------------|
|                                                   |        |        | target Smart Cities Mission fast-track procurement                                                   |
| Technical challenges in real-time data processing | Medium | High   | Scalable cloud microservices architecture; extensive simulation testing before live deployment       |
| Competition from global players                   | Medium | Medium | Emphasize India-specific AI, cost advantage, and local support; build government relationships early |
| Low camera quality at target intersections        | Medium | Medium | Develop low-light and occlusion-robust AI models; include camera upgrade advisory in pilot scope     |
| Key personnel departure                           | Low    | High   | Document all processes; cross-train team members; establish equity-based retention incentives        |



## **CHAPTER 11**

### **CONCLUSION**

The Intelligent Traffic Management Using Computer Vision project presents a technically sound, commercially viable, and socially impactful solution to one of India's most pressing urban challenges. The platform's AI-driven adaptive signal control and automatic emergency vehicle green corridor capabilities address a validated, critical gap in existing traffic management infrastructure.

The solution aligns directly with SDG 11 (Sustainable Cities and Communities), SDG 3 (Good Health and Well-being), and SDG 13 (Climate Action), contributing measurable reductions in commute times, emergency response delays, and vehicular emissions. Market research confirms a large and growing opportunity, with the Indian intelligent traffic management market projected at Rs. 4,600 crore SAM by 2028.

The SaaS business model ensures financial sustainability with a projected path to profitability by Year 2, supported by strong unit economics (CLV/CAC ratio of 5.5x, 6-month payback period). The team's combined expertise in AI/ML, system integration, and business development positions us effectively to execute the go-to-market strategy.

Future scope includes integration with autonomous vehicle communication protocols (V2X), expansion to Southeast Asian smart city markets, and development of a predictive traffic management module that optimizes signal plans 24–48 hours ahead using weather and event data. The IDL-II experience has deepened our entrepreneurial competencies and confirmed our commitment to building this venture beyond the academic context.

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## **APPENDIX I**

### **SOURCE CODE / DATA SHEET / ANY OTHER RELEVANT DATA**

Project Repository: <https://haritharan-s.github.io/Nameless-Project/>

Demo Video: <https://youtu.be/fPSPeEAfJNs>

### **Technical Architecture**

- **AI & Machine Learning:** Built with Python and TensorFlow; models specifically trained on Indian traffic patterns for superior adaptability.
- **Data Management:** PostgreSQL database handles massive real-time traffic data with high reliability and performance for analytics and reporting.
- **User Interface:** React/Node.js dashboard provides city officials with an intuitive, real-time view of traffic flow and system performance.
- **Cloud Infrastructure:** Cloud APIs enable seamless integration with existing city infrastructure, sensors, and emergency vehicle GPS systems.

## **APPENDIX II**

### **KEY PERFORMANCE INDICATORS (KPI SECTION)**

#### **1. Research Professional Practice**

- 1.1 Paper Publications: 'Intelligent Traffic Signal Optimization Using Computer Vision' — submitted to IEEE International Conference on Smart City Technologies, 2026.
- 1.2 Patent Publications: Provisional patent application filed for 'Method and System for Automatic Emergency Vehicle Green Corridor Generation Using Real-Time Video Analytics' (Application pending, 2026).
- 1.3 Project Proposals Submitted: Proposal submitted under Startup India Seed Fund Scheme (SISFS) for product development funding.
- 1.4 Community Service Projects: Free traffic congestion analysis report provided to Chennai Municipal Corporation, 2025.

#### **2. Competition / Funding**

- 2.1 Hackathon Participation: Smart India Hackathon 2025 — Shortlisted, Smart Cities theme.
- 2.2 Project-Based Competitions: Sairam Ideathon 5.0 — Participated, Urban Mobility track.
- 2.4 Grant / Funding: Seed funding application submitted to NASSCOM 10,000 Startups programme.
- 2.6 Real-Time Validation: MVP demonstrated to Chennai Traffic Department official; positive feedback received on emergency corridor feature (2025).

#### **3. Startup / Entrepreneurship**

- 3.1 Startup Registered: Company incorporation under LLP framework in progress (2026).
- 3.2 Incubation Support: Application submitted to Sri Sairam Engineering College Technology Business Incubator.

### APPENDIX III

#### JUSTIFICATION FOR POSITIVE

| S.No | Parameter     | Justification                                                                                                                                                      | Rating (1–5) |
|------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 1    | Productable   | Functional MVP developed with modular, scalable cloud-native architecture using Python/TensorFlow/React; suitable for multi-city rollout without hardware changes. | 5            |
| 2    | Opportunities | Large and rapidly growing smart city and ITS market in India, supported by the Smart Cities Mission and government digital infrastructure investment.              | 4            |
| 3    | Sustainable   | Optimizes signal timing to reduce idle time, fuel consumption, and emissions; directly aligns with SDG 11 (Sustainable Cities) and SDG 13 (Climate Action).        | 3            |
| 4    | Informative   | Provides real-time dashboards, congestion alerts, and predictive analytics for traffic planners and control-room operators.                                        | 5            |
| 5    | Technology    | Uses state-of-the-art AI/computer vision, IoT integration, and cloud-based microservices, in line with modern ITS standards.                                       | 5            |
| 6    | Innovative    | Among the first real-time adaptive traffic systems with automated emergency corridors trained specifically for Indian traffic conditions.                          | 5            |
| 7    | Viable        | Clear B2G/B2B revenue model with subscription and deployment fees; validated problem with early pilot interest from municipal stakeholders.                        | 4            |
| 8    | Ethical       | Enhances public safety, reduces ambulance delays, respects data privacy through anonymized video processing, and promotes social welfare.                          | 4            |